Tectonophysics, 40 (1977) 227—244

© Elsevier Scientific Publishing Company, Amsterdam — Printed in The Netherlands

DYKE MAGNETIZATION, MAGNETOSTRATIGRAPHY AND UPPER-CRUSTAL STRUCTURE IN THE REYDARFJORDUR AREA OF EASTERN ICELAND

J.D.A. PIPER [1], M.G. FOWLER [1] and I.L. GIBSON [2]

[1] *Sub-department of Geophysics, Oliver Lodge Laboratory, Liverpool L69 3BX (England)*
[2] *Department of Geology, Bedford College, London NW1 4NS (England)*

(Submitted May 13, 1976; revised version accepted October 12, 1976)

ABSTRACT

Piper, J.D.A., Fowler, M.G. and Gibson, I.L., 1977. Dyke magnetization, magnetostratigraphy and upper-crustal structure in the Reydarfjördur area of eastern Iceland. Tectonophysics, 40: 227—244.

A dyke swarm approximately 12 m.y. old and comprising 103 basaltic dykes was sampled during a palaeomagnetic investigation of the Reydarfjördur area, eastern Iceland. Of 100 dykes yielding statistically significant palaeomagnetic directions 36 are normal, 43 reversed and 21 are intermediate; 5% of the dykes are olivine-phyric. The magnetic stratigraphy of the lava pile in this area is partially known from earlier palaeomagnetic studies, and six additional sections measured by field magnetometer and a palaeomagnetic section of 40 flows (20 reversed, 16 normal, and 4 intermediate) are reported here. These sections can be correlated by means of petrologic horizons (silicic tuffs, olivine basalts and porphyritic units) and exhibit stratigraphic thickening towards the centre of the Reydarfjördur dyke swarm. The portion of the basalt pile cut by the dyke swarm can be related both petrologically and palaeomagnetically to the dykes, and the number of exposed dykes is found to be of the same order, but smaller than, the number of lava flows in that comparable portion of the lava pile. This implies that in general the exposed dyke swarm is related to the exposed lava pile which is unlikely to extend much further downdip than the known extension updip. The lava pile in Iceland evolves as large lenticular units formed at eruptive axes, and distributed en échelon along the length of the axial zone with spacings comparable to the thickness of the lithosphere.

INTRODUCTION

Bodvarsson and Walker (1964) and Walker (1965) were the first to recognize that Iceland has evolved by crustal growth in a manner analogous to sea-floor spreading. However, as Pálmason and Saemundsson (1974) note, the anomalously thick Icelandic crust, the widespread distribution of volcanism across more than one volcanic zone, and the development of major silicic/thermal centres in Iceland all suggest that only limited comparisons can be made with typical oceanic crust.

In eastern Iceland a thick pile of subaerial lavas, dips at low angles (12° or

less) systematically westwards towards the present neovolcanic zone. Recent K—Ar (Watkins et al., 1976) and Ar_{39-40} (Ross and Mussett, 1976) age work demonstrates that this pile has grown by essentially uninterrupted volcanism during the last 13 m.y. from a broad neovolcanic zone or zones. However, the addition of material to the pile at any one point along the neovolcanic zone has probably not been regular because the present centres of silicic and thermal activity are strictly localized, and individual groups of older lavas are observed to change their thickness laterally. Dykes, the exposed feeders of the lavas, are also not uniformly distributed, but are concentrated into swarms (Walker, 1959, 1960; Gibson, 1969) comprising up to about 15% of the crust and separating zones in which dykes are virtually absent.

In this paper we report a palaeomagnetic study of a dyke swarm cutting the lava pile near its exposed base in the Reydarfjördur area of eastern Iceland; we also report palaeomagnetic and magnetic-stratigraphic information from the lava pile and use the combined petrologic and palaeomagnetic data to relate the dykes to the extrusive lava pile. This new information yields some further information on the growth of the upper crust in Iceland.

PALAEOMAGNETISM OF THE DYKES

The Reydarfjördur dyke swarm in eastern Iceland was mapped initially by Walker (1959) and the inset diagram of Fig. 1 is taken from this reference; data for the southern extension of the swarm are given in Gibson et al (1966, fig. 22). These studies showed that most dykes trend north—south and are approximately perpendicular to the westward-dipping lavas with inclinations of 5—10°. Both dykes and lavas appear to have been progressively tilted sub-

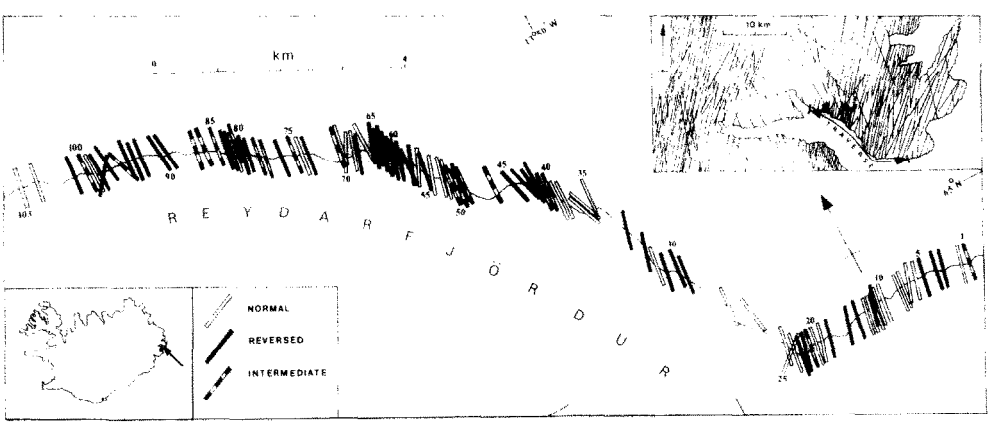


Fig. 1. Distribution and magnetic polarities of dykes in the Reydarfjördur area, eastern Iceland. The inset diagrams illustrate the regional location of the dyke swarm and the distribution of dykes in the Reydarfjördur area after Walker (1959). The numbers refer to sampling site numbers for which palaeomagnetic data are listed in Table I.

sequent to their formation probably during accretion of the lava pile, and the amount of dip generally increases with depth in the pile (Walker, 1965; Gibson and Piper, 1972).

The dyke swarm is most dense at sea level and diminishes rapidly with increasing altitude in the exposed lava pile. This decrease has been demonstrated quantitatively by detailed measurement in the Berufjördur region 35 km to the south of Reydarfjördur (Walker, 1960), but we only have qualitative information in this area (Walker, 1959, fig. 6; Gibson et al., 1966, figs. 22 and 24). The dykes are most abundant along Reydarfjördur near the outcrop of the silicic lavas of the Séllatur acid group of Walker (1959) and some highly propylitized tholeiitic lavas, where the main outcrop of the swarm is associated with extensive hydrothermal alteration of the pile; this effect is superimposed on the zones of zeolitization produced by burial metamorphism (Walker, 1960, 1964).

Altogether 103 basaltic dykes exposed along the north shore of Reydarfjördur (Fig. 1) were sampled with a two-stroke portable drill (Doell and Cox, 1967) and from three to five cores were taken from each dyke, and oriented in situ using a sun or magnetic compass. The sampled dykes range from 0.5 to 15 metres in thickness and average 3 metres.

In the laboratory magnetic moments were measured with a parastatic magnetometer before and after progressive demagnetization in steps of 50 or 100 oersteds (Oe) up to 250 or 300 Oe. Directions of magnetization were averaged over the most stable range of demagnetization behaviour (typically 100—200 Oe) and directions from each sample combined to give the site means listed in Table I. All directions of magnetization were corrected for a tectonic dip averaging 8° in a direction 270°E. The minimum number of samples measured from each site was 3 and where a single sample yielded a direction of magnetization widely removed from the other two it was rejected. No significant grouping at the 95% confidence level was found from three of the dykes.

After a.f. cleaning most site mean directions of magnetization cluster around the present or normal geomagnetic field direction (declination = 0°, inclination = +77°) or the opposite or reversed direction (declination = 180°, inclination = —77°, Fig. 2) although an appreciable number do not lie near either of these directions. To determine the polarity of a dyke, we define normal and reversed modes as those yielding virtual geomagnetic poles or antipoles falling within 40° of the present mean geographic pole; other poles are classed here as intermediate. The percentage of dykes from this study yielding intermediate directions (21%) is higher than in the palaeomagnetic study of the related lava sequence (Dagley et al., 1967, 7%), but since some of these sites have poor groupings (Table I) we prefer to exclude them from further consideration and consider only those dykes with established normal or reversed polarities.

The distribution of dyke polarities is illustrated in Fig. 1 where it is seen that the dykes are not in general concentrated into swarms of a single polar-

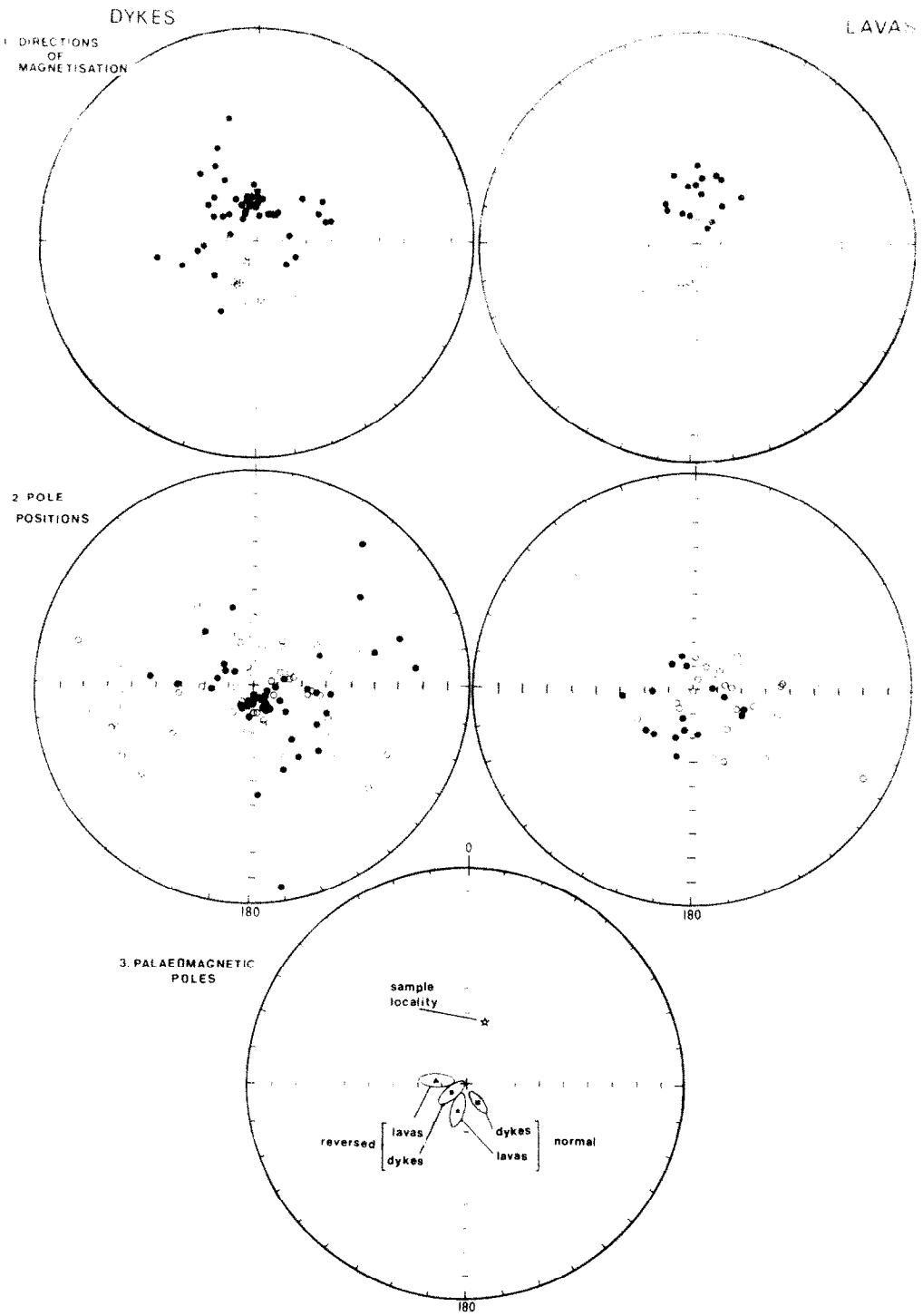


Fig. 2. Directions of magnetization and virtual geomagnetic poles from dykes and lavas of the Reydarfjördur area, eastern Iceland (Tables I to III); closed symbols are lower-hemisphere and open symbols are upper-hemisphere plots. The lower plot illustrates palaeomagnetic poles from normal dykes and lavas and antipoles from reversed dykes and lavas together with 95% confidence ovals. Polar stereographic projections.

TABLE I

Site mean palaeomagnetic statistics for dykes from the Reydarfjördur area of eastern Iceland

Dyke No.	Type	Polarity	*D*	*I*	*N*	*R*
1	Th	I	260.8	48.0	2	1.89
2	Th	N	319.1	53.1	3	2.96
3	OB	R	172.4	−73.3	4	3.86
4	OB	R	171.3	−59.3	4	3.92
5	Th	R	163.9	−77.7	3	2.98
6	Th	N	358.8	74.5	3	2.99
7	Th	N	7.9	79.5	4	3.95
8	Th	N	346.0	71.5	4	3.97
9	Th	N	21.9	77.9	5	4.94
10	Th (P)	N	31.9	76.8	4	3.98
11	Th	N	348.0	73.3	4	3.95
12	Th (P)	N	58.1	58.1	4	3.93
13	Th	R	216.9	−83.2	5	4.99
14	Th (P)	N	331.7	70.0	4	3.98
15	Th	N	328.5	78.3	4	3.99
16	Th	R	202.1	−76.6	4	3.97
17	Th	R	196.6	−71.0	4	3.99
18	Th	R	221.6	−69;3	3	2.98
19	Th (P)	N	335.2	73.3	4	3.95
20	Th	N	345.4	75.4	4	3.98
21	Th	I	334.9	−32.5	3	2.86
22	Th	R	195.6	−81.8	2	1.99
23	Th	R	134.0	−81.5	3	2.98
24	Th	N	359.1	71.7	4	3.99
25	Th	N	329.9	53.3	3	2.99
26	Th	N	331.7	60.4	3	2.99
27	Th	N	3.0	70.0	3	2.97
28	Th (P)	N	354.4	70.6	3	2.99
29	Th	R	21.9	−71.7	3	2.99
30	Th (P)	R	203.2	−69.7	3	3.99
31	Th	R	163.7	−63.1	2	1.97
32	Th	N	315.9	64.4	3	2.85
33	Th	R	143.4	−61.1	4	3.84
34	Th	R	159.8	−83.1	3	2.99
35	Th	N	281.6	77.8	3	2.94
36	Th	N	299.5	69.2	3	2.99
37	Th	N	46.1	64.4	3	2.99
38	Th	N	34.7	77.2	3	2.99
39	Th	N	31.6	74.1	4	3.99
40	Th (P)	N	74.5	−58.3	3	2.95
41	Th (P)	I	75.4	−59.5	2	1.99
42	Th	R	80.0	−76.4	3	2.99
43	Th	R	113.5	−73.4	3	2.98
44	Th	R	130.7	−75.2	2	1.96
45	Th (P)	R	144.2	−49.6	2	1.74
46	Th	I	230.9	67.7	3	2.99
47	OB (P)	I	65.8	−62.3	4	3.99

TABLE I (continued)

Site mean palaeomagnetic statistics for dykes from the Reydarfjördur area of eastern Iceland

Dyke No.	Type	Polarity	*D*	*I*	*N*	*R*
48	Th	R	146.6	−62.5	2	1.82
49	Th	I	91.4	−55.5	3	2.85
50	Th	I	101.6	−53.1	3	2.97
51	Th	I	268.7	−49.1	3	2.71
52	Th	N	358.3	74.3	3	2.99
53	Th	N	355.5	65.8	3	2.99
54	Th	N	338.3	77.1	3	2.99
55	Th	R	149.2	−68.2	3	2.99
56	Th (P)	N	305.7	72.4	3	2.99
57	Th (P)	R	172.5	−63.6	2	1.99
58	Th	R	253.3	−70.4	3	2.98
59	Th	R	207.5	−80.2	3	2.98
60	Th	R	206.6	−70.2	3	2.98
61	Th (P)	I	60.9	−62.4	2	1.99
62	Th (P)	R	220.1	−61.2	2	1.98
63	Th	R	191.7	−73.8	3	2.99
64	Th	R	252.2	−74.7	3	2.93
65	Th (P)	R	226.1	−68.1	3	2.99
66	OB	N	347.5	72.8	3	2.97
67	Th	R	189.4	−62.6	3	2.99
68	Th (P)	N	355.0	75.0	3	2.99
69	Th		not significant			
70	Th	N	5.0	71.9	3	2.99
71	Th (P)	N	307.2	65.3	4	3.96
72	Th (P)	I	261.5	64.4	3	2.97
73	Th (P)	R	242.7	−68.6	4	3.85
74	Th	N	336.4	47.8	3	2.99
75	Th	I	90.5	−59.8	2	1.99
76	Th	R	176.0	−65.7	3	2.94
77	Th	I	233.2	−60.4	3	2.99
78	Th	I	230.3	−67.0	3	2.96
79	Th	R	189.2	−63.8	3	2.97
80	Th	R	208.6	71.2	3	2.96
81	Th	I	252.0	57.1	4	3.98
82	—	N	346.2	71.6	4	3.91
83	Th	I	79.4	−49.1	3	2.98
84	Th	R	205.6	−70.9	2	1.97
85	—	I	96.5	−26.9	3	2.89
86	Th		not significant			
87	Th	I	346.5	36.5	3	2.90
88	Th	I	207.6	57.6	2	1.99
89	Th	R	160.4	−53.8	3	2.65
90	Th	R	181.0	−65.2	2	1.99
91	Th	R	173.6	−79.6	3	2.99
92	Th	R	211.6	−83.2	3	2.99
93	Th	R	273.7	−79.9	3	2.98
94	Th	I	224.4	−44.7	3	2.84

TABLE I (continued)

Site mean palaeomagnetic statistics for dykes from the Reydarfjördur area of eastern Iceland

Dyke No.	Type	Polarity	D	I	N	R
95	Th		not significant			
96	Th	R	272.9	−53.7	3	2.56
97	Th	R	187.2	−54.2	4	3.86
98	Th	R	274.3	−77.2	2	1.99
99	Th	N	312.6	73.5	3	3.97
100	Thq	I	267.7	−67.3	3	2.99
101	Th (P)	R	165.3	−65.6	3	2.98
102	OB	N	358.2	68.6	3	2.98
103	Th	N	348.0	74.7	4	3.99

D and I are the declination and inclination of the direction of magnetization derived from N samples with R as the magnitude of the resultant vector. Th = tholeiite, OB = olivine basalt. In the polarity column N = normal, R = reversed, I = intermediate. (P) = porphyritic lava.

ity, but both polarities are found throughout the traverse. The implications of this observation and the relationship of the dykes to the lavas are discussed in the section on Relationship between the dykes and lavas.

Directions of magnetization and pole positions are illustrated in Fig. 2 and combined palaeomagnetic statistics for all dykes are listed in Table II. Basaltic volcanism in Iceland produces both olivine basalts and tholeiitic basalts and in eastern Iceland the latter comprise rather more of the total sequence than the former. Although there is a broad region of overlap between the two divisions, they can often be classified in the field on the basis of their physical characteristics (Walker, 1959; Saemundsson, 1967) and zeolite assemblages (Walker, 1960). To give a positive identification thin sections were

TABLE II

Group mean palaeomagnetic statistics for dykes and lavas, eastern Iceland

	N	D	I	R	α_{95}	Pole position		$d\Psi$	$d\chi$
						°N	°E		
Reversed dykes	43	8.9	74.2	41.51	4.2	83.9	120.5	6.8	7.6
Normal dykes	36	347.5	72.7	35.17	3.7	80.9	212.6	5.9	6.6
All dykes	79	358.6	73.8	76.56	2.7	84.8	173.7	4.6	5.1
Reversed lavas	20	25.1	74.1	19.65	4.5	77.7	86.8	7.3	8.1
Normal lavas	16	00.9	69.6	15.69	5.4	78.4	163.2	7.9	9.2
All lavas	36	12.9	72.5	35.24	3.5	80.5	119.8	5.6	6.3

D and I are the declination and inclination of the direction of magnetization derived from N sites with R the magnitude of the resultant vector; α_{95} = radius of 95% confidence circle of the mean direction; $d\Psi$ and $d\chi$ = semi-angles of 95% confidence of pole position.

examined to classify all the dykes, and this information is included in Table 1 where it is seen that the bulk of the dykes are tholeiites.

MAGNETIC STRATIGRAPHY OF THE LAVA PILE

The geology of the Reydarfjördur area has been studied in detail by Walker (1959) and Gibson et al. (1966), and the maps of these workers illustrate the regional outcrop of the principle stratigraphic divisions. In general silicic tuffs represent the most widely spread products of any igneous episode and are accordingly the best stratigraphic marker horizons. Olivine-rich basalt flows and feldspar-phyric flows tend to occur in groups which can be followed for long distances along the strike, and are also useful marker horizons, although representing as they do, a number of flows, they cannot be regarded as true time planes like the silicic tuffs. In general silicic and intermediate lava flows are the most restricted units in the stratigraphy, and in Reydarfjördur their outcrop is largely confined to the north and south shores of the fjörd close to the axis of the dyke swarm.

Information on the magnetic stratigraphy in this area is derived from the palaeomagnetic traverse of Dagley et al. (1967) which included a number of overlapping stratigraphically controlled sections in the northern part of

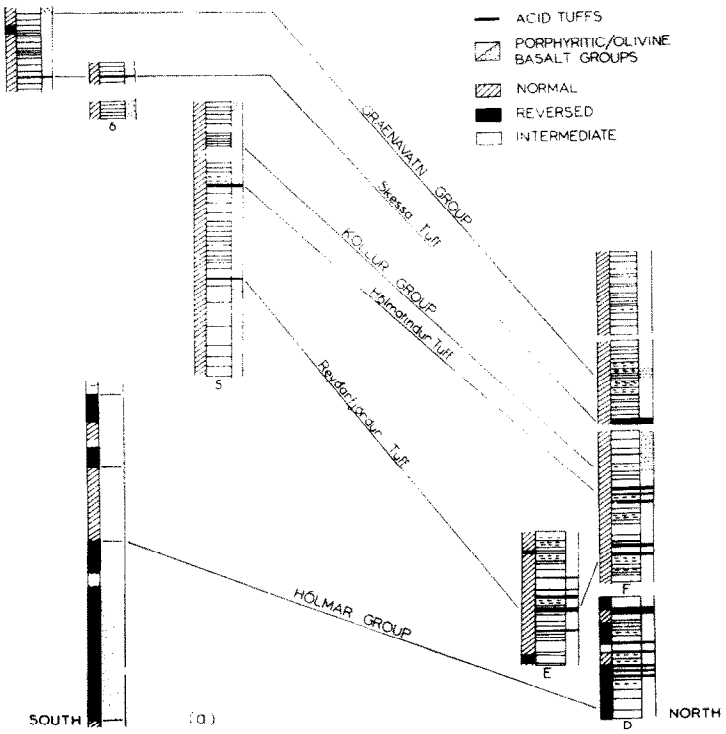


Fig. 3a.

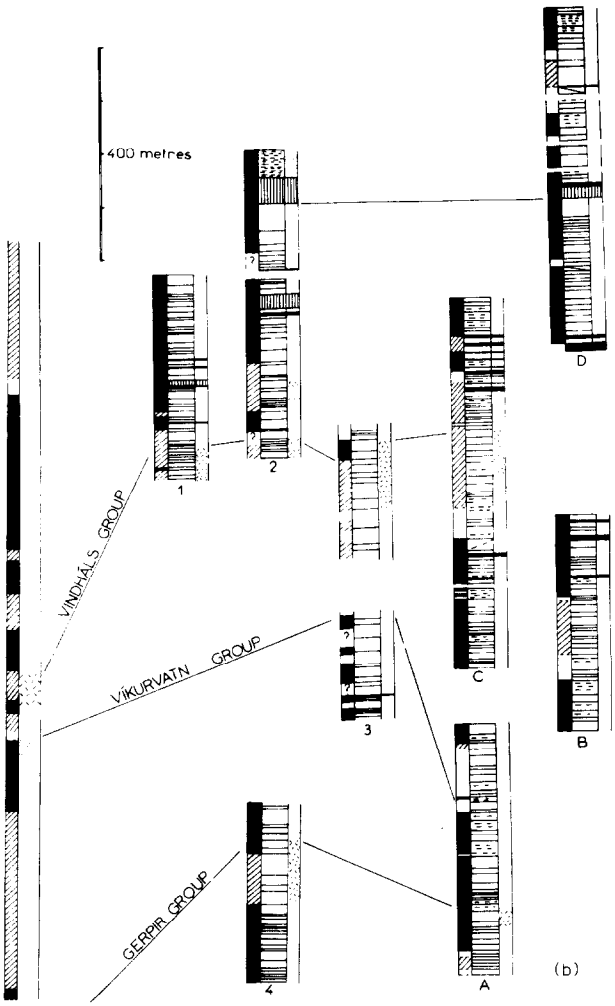


Fig. 3. Summary of present information relating to the magnetic stratigraphy of (a) the upper part and (b) the lower part of the lava pile in the Reydarfjördur area of eastern Iceland using a constant vertical scale for stratigraphic thickness. The location of these sections is given in Fig. 4. Sections on the right-hand sides are from Dagley et al. (1967) and are dip-corrected, and the stratigraphic column derived from palaeomagnetic results from Reydarfjördur (Fig. 4 and Table III) is calculated for a mean dip of 8° in a direction 270°E. The six sections measured in the field are not dip-corrected like the remaining sections; this dip is low and varies from about 8° at the base of the sections to about 4° at the top. The principle stratigraphic marker horizons are correlated between the sections, and are described by Walker (1959) and Gibson et al. (1966).

the Reydarfjördur area and has since been described in more detail with additional age data by Watkins et al. (1976). Additional information to be described here comes from a palaeomagnetic traverse along the two parts of

Reydarfjördur, and six stratigraphically controlled and measured sections in which magnetic polarities were measured on hand samples in the field using a fluxgate magnetometer.

All of this information is assembled in Fig. 3 where marker horizons are connected to illustrate the lateral changes in thickness of the lava pile between the measured sections. Although the field magnetic stratigraphy is based on exhaustive tests on samples from all parts of the lava flows, it is always possible that large components of secondary magnetization are present throughout the thickness of some flows, and the indicated field magnetic stratigraphy accordingly has lower reliability. However, as Fig. 3 illustrates, agreement between overlapping stratigraphic sections based on field and laboratory studies is good, and the sections measured with a fluxgate magnetometer provide useful supplementary information.

The discontinuous palaeomagnetic traverse along the north shore of Reydarfjördur comprises 40 flows (Fig. 4) sampled and treated in the same way as the dykes, and the palaeomagnetic statistics and petrological data are given in Table III where the lavas are classed as normal or reversed using the same criterion as that specified for dykes. The vertical palaeomagnetic stratigraphy shown in Fig. 3 is calculated using a mean dip of 8° to the west, and as only some flows were sampled in this traverse, certain magnetic reversals can be expected to have missed although the petrologic marker horizons do permit this long section to be correlated with the remaining traverses.

The most striking feature of Fig. 3 is the southward thickening of the central part of the stratigraphic succession as it is traced towards Reydarfjördur. This thickening is also apparent from the sea-level outcrop of certain major stratigraphic units (Walker, 1959 and Fig. 4). Along Reydarfjördur this expanded section includes over 500 metres of tholeiite flows and silicic volcanics of the Séllatur acid Group. This succession thins dramatically northwards and in the Nordfjördur region, 12 km to the north, only a condensed sequence including a series of closely spaced silicic tuffs is observed (Fig. 3) totalling about 200 metres. Although, because of the difficulty of precisely correcting for updip thinning of the lava succession, it is impossible to give a precise quantitative estimate from the data of Fig. 3, it would appear that the lava pile between the Vindháls Porphyritic Group and the Hólmar Olivine Basalt Group thins by approximately 50% in 15 km northwards along the strike. In the opposite sense, we observe that the succession between the Vindháls Olivine Basalt Group and the Vikurvatn Group thickens northwards where it includes a number of silicic tuffs in the Nordfjördur region. This thickening occurs at the stratigraphic level of the Lodmundarfjördur silicic centre with a main outcrop approximately 25 km further to the north. Above and below the rocks related to the Reydarfjördur silicic centre, the tholeiite and olivine basalt divisions do not show appreciable thickness variations (Fig. 3) and slight variations in thickness are an order of magnitude smaller than the thickness variations in the portion of the lavas involved in the Reydarfjördur silicic centre; they cannot effectively be resolved from differences

TABLE III

Palaeomagnetic statistics for lava flows, Reydarfjördur area, eastern Iceland: the sites are listed in stratigraphic order (Fig. 4)

Site No.	Type	Polarity	D	I	N	R
1	OB	R	161.4	−80.8	4	3.95
2	OB	N	30.6	82.2	3	2.99
3	OB	N	34.3	79.2	2	1.99
4	OB	N	14.5	60.9	3	2.98
5	T	N	42.2	63.9	4	3.98
6	T	N	3.9	61.9	4	3.91
7	T	R	202.8	−64.6	3	2.94
8	OB	R	228.0	−60.7	4	3.79
9	OB	I	123.9	−15.2	2	1.99
10	T	N	30.4	71.3	3	2.97
11	T	R	224.0	74.7	4	3.98
12	P	N	341.4	77.5	3	2.97
13	T	R	107.8	−79.4	2	1.99
14	T	R	275.1	−80.3	3	2.99
15	T	N	18.9	60.6	3	2.75
16	T	R	235.4	−78.7	2	1.99
17	T	N	319.3	70.6	4	3.55
18	T	R	228.3	−70.7	2	1.99
19	T	R	242.0	−65.6	4	3.98
20	T	R	207.2	−71.9	3	2.99
21	T	R	166.1	−81.5	3	2.98
22	T	R	183.4	−70.8	3	2.83
23	T	I	235.9	−47.5	3	2.78
24	T	N	2.1	59.3	3	2.63
25	T	N	350.2	64.6	3	2.97
26	T	N	321.7	69.6	3	2.95
27	T	N	332.7	75.8	4	3.99
28	OB	R	181.7	−74.2	4	3.97
29	OB	R	174.7	−68.5	3	2.78
30	OB	R	198.6	−63.1	3	2.98
31	OB	I	276.5	−28.6	4	3.72
32	OB	R	202.0	−72.7	3	2.95.
33	T	R	179.5	−77.9	3	2.99
34	T	N	359.3	65.1	4	3.91
35	T	N	359.7	56.1	3	2.96
36	OB	R	227.1	−60.3	3	2.85
37	OB	I	267.0	−70.3	3	2.98
38	OB	N	340.5	59.1	3	2.99
39	OB	R	190.9	−72.7	2	1.99
40	OB	R	193.5	−70.3	3	2.98

D and I are the declination and inclination of the remanence vector derived from N samples with a resultant vector equal to R; in the polarity column N = normal, R = reversed, and I = intermediate. Th = tholeiite, OB = olivine basalt.

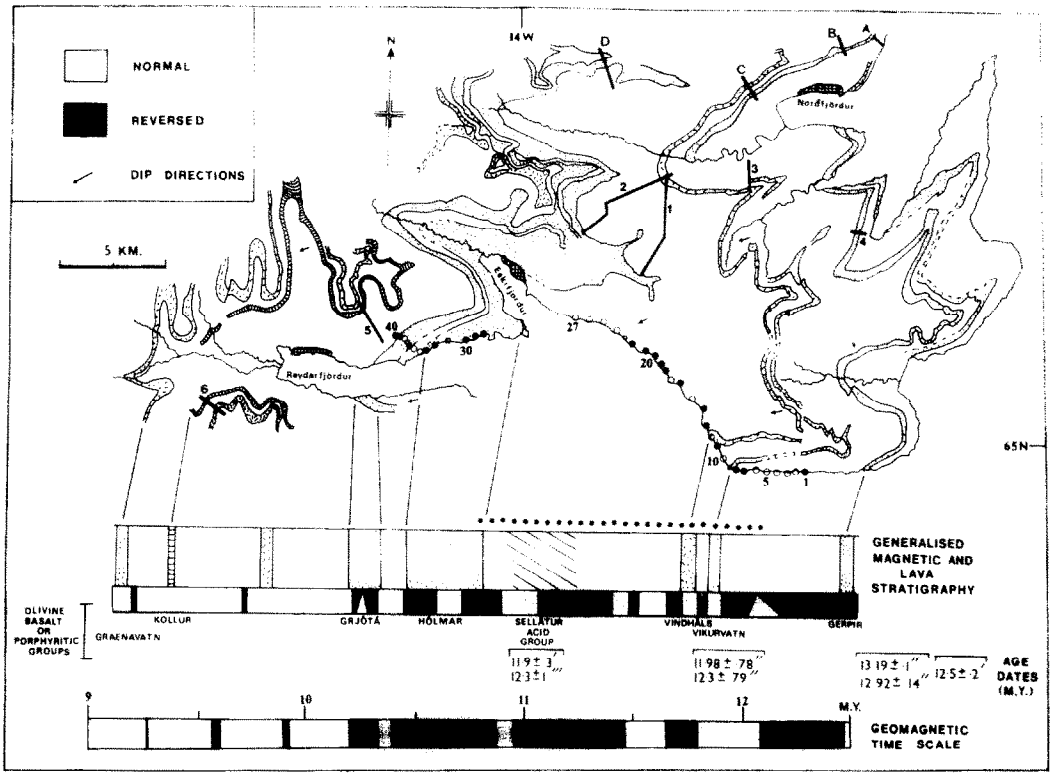


Fig. 4. Outcrop of major petrologic divisions of the lava pile in eastern Iceland after Walker (1959). The sections for which palaeomagnetic-stratigraphic information is given in Fig. 3 are indicated together with palaeomagnetic sampling sites in lava flows (closed circles are reversed, open circles are normal and squares are intermediate polarities (Table III)). Two magnetic-stratigraphic columns are given. The first is a generalized stratigraphy from the palaeomagnetic information of Fig. 3 with the principle petrologic divisions indicated together with the portion of the lava stratigraphy (line of closed dots) fed by the dyke swarm illustrated in Fig. 1. The second column is the geomagnetic time scale for part of Miocene times deduced from stacked marine magnetic anomalies by Blakely (1974); the time period covered by this time scale is approximately the same as that covered by the lava stratigraphy in this part of Iceland. K—Ar and Ar_{39}—$_{40}$ ages relating to the lava stratigraphy in this area are: 'Moorbath et al. (1968), ''Ross and Mussett (1976), '''Charlton (1973).

resulting from variation in altitude within the lava pile at which the observations were made.

Changes in thickness of the Icelandic lava pile both along and at right angles to the strike have been mapped by Gibson (1966, 1969) in stratigraphic subdivisions within the Reydarfjördur centre, and by Gibson and Piper (1972) in palaeomagnetically defined subdivisions of the lava pile in S.W. Iceland. As in the case recognized here, thickening takes place towards the silicic centres and (where exposed) towards the regional dyke swarms.

This leads logically to the conclusion that volcanism was concentrated at these points in the crust, leading to the growth of lenticular lava units (Gibson and Piper, 1972).

RELATIONSHIP BETWEEN THE DYKES AND LAVAS

It is generally believed that the dykes in Iceland have fed the older lava flows because, with the exception of a few minor plug-like intrusions, they are the only visible feeders (Walker, 1959). It is also probable that the dykes exposed at high crustal levels are the product of activity under the highest hydrostatic pressures and are therefore those dykes most likely to have produced the subaerial flows. However, not all the dykes will be exposed and many more may cut and feed the concealed down-dip section at the lava pile. The palaeomagnetic and petrologic information of Table I provides some more definitive evidence with which to relate the dykes to the lavas.

The sampled dykes are emplaced into a predominantly tholeiitic sequence of flows between the Gerpir Porphyritic Group and the Hólmar Olivine Basalt Group. In the west the traverse stops short of the upper part of the Séllatur Acid Group, and as very few of the dykes exposed at sea level here are observed at appreciably higher altitudes cutting younger stratigraphic units we infer that they are unlikely to have fed lavas above the base of the Hólmar Olivine Basalt Group (Fig. 4). The portion of the stratigraphy fed by the sampled dykes is therefore predominantly tholeiitic, and embraced by the dotted line in Fig. 4. This is consistent with the petrology of the dykes, all but five of which are tholeiites. The only olivine basalt unit mapped in this succession is the thin Vikruvatn Olivine Basalt Group of reversed and intermediate magnetization which could have been fed by the two reversely magnetized olivine basalt dykes 3 and 4 (Fig. 5). The other three olivine basalt dykes sampled occur above this unit and can only be related to the overlying

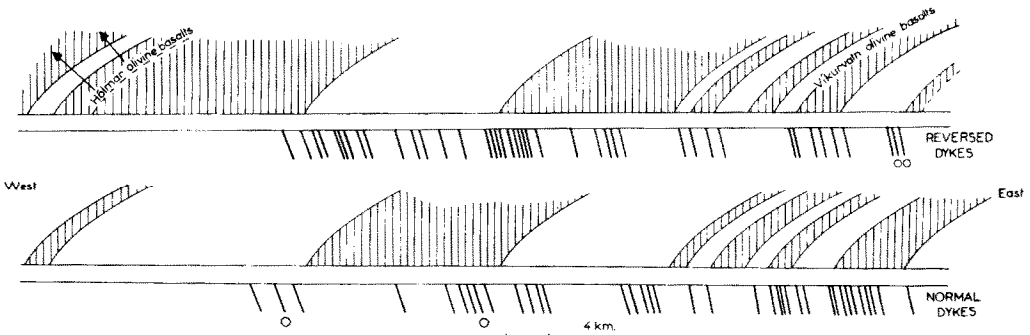


Fig. 5. Schematic cross-section through the lava pile along Reydarfjördur illustrating the outcrop of the dykes and the portion of the lava stratigraphy which they may have fed (vertical shading); The magnetic stratigraphy of the lava pile is deduced from the information of Figs. 3 and 4. 0 = olivine basalt dykes, the remainder are tholeiites.

Hólmar Olivine Basalt Group. The remaining (tholeiite) dykes comprise 34 with normal magnetizations, 41 with reversed magnetizations and 20 with intermediate magnetizations. The ratio of normal to reverse polarities is approximately the same as the ratio of normal to reversed polarities of the tholeiite lavas in what we interpret as the related part of the succession. The high percentage of intermediate magnetizations is consistent with the frequent reversals (at least 9) in the lava succession.

We can also estimate the approximate number of lavas comprising the sequence fed by these dykes from the information of Fig. 3 although we note that the measured successions in which the number of lavas have been counted in Fig. 3 are condensed relative to the succession at Reydarfjördur. The 36 normal and 43 reversed dykes of our study appear to relate to a lava succession comprising at least 38 normal and 60 reversed flows. From this observation we conclude that the dykes exposed at sea level *may* have fed the bulk of the exposed lava pile although at least some of the feeders must be hidden downdip.

AGE OF THE DYKES AND LAVA SUCCESSION

Geochronological information relating to the lava pile in eastern Iceland is summarized in Fig. 4 and is derived from K—Ar ages of Moorbath et al. (1968) and Charlton (1973), and Ar_{39}—Ar_{40} ages of Ross and Mussett (1976). Age estimates of the base of the stratigraphic succession range from 12.5 ± 0.2 to 13.2 ± 0.1 m.y., while the ages derived from near the base of the succession intruded by the dyke swarm (11.9 and 12.3 m.y.) are not significantly different from those derived from the higher part of the sequence probably fed by the dykes studied here (11.9 and 12.3 m.y.). It is possible therefore that the dyke swarm and the lava sequence represent a time span considerably less than 1 m.y. between 11 and 12 m.y. ago. More reversals are recognized in this basalt succession than in marine magnetic profiles covering the same interval (Heirtzler et al., 1968; Blakely, 1974) although it is of course possible that some of the reversals recognized in eastern Iceland are only local excursions of the geomagnetic field. As a result it is only possible to correlate the magnetic profiles with the geomagnetic time scale in a rather general way. We observe that the lava sequence above the Hólmar olivine basalt group and below the Thingmuli silicic centre dated 9.5 ± 0.6 m.y. by Moorbath et al. (1968) is predominantly normally magnetized. It correlates with marine magnetic anomaly 5, the interval between 8.71 m.y. and 10.21 m.y. when the earth's field was predominantly normally magnetized (Blakely, 1974, and Fig. 4). As also observed by Blakely, we find that at least two short reversed events are represented in this predominantly normal sequence (Fig. 3). The contrast between the succession of mixed polarity and the higher succession of normal polarity is expressed in the aeromagnetic anomalies over this area (Serson et al., 1968), the main features of which can be

followed for over 60 km along the strike and approximately explained in terms of the magnetic stratigraphy of the lava pile (Piper, 1973a).

INTERPRETATION OF THE PALAEOMAGNETIC POLES

Of the one hundred dykes yielding significant groups of directions 36 have normal, 43 have reversed and 21 have intermediate directions; the lava collection yielded 40 significant sites with 20 reversed, 16 normal and 4 intermediate. The directions have been combined to yield normal and reversed poles plotted in Fig. 2. All of these poles are significantly different at the 95% confidence level and far-sided, with respect to the geographic pole. This is also true for other palaeomagnetic poles from the eastern Icelandic lava succession (Wilson, 1970). This phenomenon was examined in a global context by Wilson (1970,1971) who showed that the time-average geomagnetic field can produce this effect if the field source is a dipole offset to the north of the equatorial plane.

We also observe that the two poles from reversely magnetized lavas and dykes are systematically displaced from the two poles from normally magnetized lavas and dykes; this relationship does not apply to the time-averaged results of the eastern Icelandic lava sequence of Watkins et al. (1976) covering the interval 12—2 m.y. and may be a short-term effect only. The reversed poles are not offset to the same extent as the normal poles although both are right-handed with respect to an observer at the sampling locality. They are similar to the Upper Tertiary and Quaternary results of Opdyke and Henry (1969) and Khramov and Sholpo (1967) analyzed by Wilson (1970, 1972) in that the poles are right-handed, but in this case the reversed poles would result from a dipole less offset from the equatorial plane than that responsible for the normal poles. We can clearly only interpret this observation in a local context here, but taken at this face value it implies that the asymmetry of the dipole source was also present ca. 12 m.y. ago although in an opposite sense to that in more recent times.

IMPLICATIONS TO MODELS FOR CRUSTAL STRUCTURE IN ICELAND

The geological implications of this study are twofold. Firstly further evidence has been provided to show that portions of the basaltic lava pile thicken towards the feeding dyke swarm; at Reydarfjördur this thickening is largely accounted for by an increasing proportion of olivine-poor, tholeiitic basalts. Secondly, the palaeomagnetic and petrologic information from the dykes is consistent with them having fed the lavas, and the dykes exposed at sea level (or about 1.5 km below the original top of the lava pile) are sufficient in number to have fed the bulk of the exposed lava pile. This latter observation is pertinent to models for the structure of the crust in eastern Iceland because it implies that dyke swarms within the top 25% of the crust and accounting for no more than about 10% of the total crustal extension can

explain the extrusive lava pile which is everywhere the dominant exposed element of the crust.

Models for the crustal structure in Iceland achieve a 100% spreading by dyke injection at the base of the lava pile. Some assume that the base of pile coincides with the base of crustal seismic layer 2 (Gibson and Piper, 1972) which lies between 2 and 5 km below the surface over most of Iceland (Pálmason, 1973), while others assume that segments of the lava stratigraphy occur down to the base of the crust (Walker, 1965; Pálmason, 1973). Neither is entirely in accord with observation (Pálmason and Saemundsson, 1974), and the complete explanation may involve both minor intrusions and metamorphic zonation.

Pálmason (1973) assumes a uniform zone of dyke injection and a related zone of continuous subsidence of the neovolcanic zones and using a quantitative model shows that continuous crustal growth can achieve a progressive dip towards these zones with the lava units asymptotic towards the upper surface of the pile. This model does not accommodate the distribution of the dykes in swarms or the lenticular nature of the pile, although this is incorporated in the qualitative model of Gibson and Piper (1972). The latter model suggests that the downdip extension of the pile is unlikely to be very much greater than the visible updip portion and consistent with the conclusion of the penultimate paragraph.

A model assuming lenticular growth of the pile is better able to reconcile the structure of the older Tertiary areas with that of the present neovolcanic zones. In this model, the lava pile is built up of lenses of extrusive volcanics each centred on an area of silicic volcanics and each with a life span of the order of 0.5—1 m.y. (Gibson and Piper, 1972). Subsequent centres appear to remain intact as they move away from the neovolcanic zone with new centres initiated en échelon with respect to the older ones. This is precisely the situation observed along the present neovolcanic zones (Piper, 1973b; Pálmason and Saemundsson, 1974; Walker, 1975) where silicic and thermal areas are distributed en échelon at intervals of 30—100 km and are also the loci of most of the basaltic volcanism; they appear to be the contemporary analogues of the lenses in the Tertiary lava pile.

Dykes, either as feeders to lava flows or as intrusive episodes which did not reach the surface, form a integral element of the oceanic crust and magnetic strip anomalies can conveniently be fitted by vertically sided blocks (Vine, 1966; Heirtzler et al., 1968) which might in real terms represent groups of dykes of one polarity. There is, however, considerable uncertainty both in Iceland and in truly oceanic areas as to the contribution of dyke swarms to oceanic magnetic anomalies. Dykes will also not be progressively intruded along a single plane and their injection is more likely to be concentrated about a narrow tensional zone, the width of which can be estimated from the degree of smoothing of surface anomalies (Matthews and Bath, 1967; Harrison, 1968; Piper, 1973a). The remoteness of the source renders such estimates rather crude and deep-towed magnetometer profiles are pro-

viding more realistic information on this point (eg. Larson and Speiss, 1969; Klitgord et al., 1975).

In Iceland the observed distribution of volcanism is much greater than elsewhere on the mid-ocean ridge system (eg. Walker, 1975) and may be a consequence of the much greater production of basalt in this area as evidenced by a thicker crust. We therefore regard the study of eastern Icelandic dykes described in this paper as of more local significance, and the extent to which observations made in Iceland can be extrapolated to the remainder of the ocean crust remains uncertain. However, it seems likely that the observed lenticular growth of the crust in Iceland is part of a universal volcanic phenomenon. It is now widely recognized that central volcanoes develop with an optimum spacing related to the thickness of the lithospheric plate (eg. Marsh and Carmichael, 1974; Vogt, 1974); and we suspect that the spacing of lenses of basaltic volcanism is controlled in a comparable way.

ACKNOWLEDGEMENTS

The fieldwork for this study was undertaken during two expeditions run under the auspices of Educational Expeditions International and we are very grateful to the Icelandic Research Council for permission to work in Iceland. We are also very grateful to Dr. Lyle Henage and student members of the expedition who helped with the collection of the palaeomagnetic samples. The work would not have been possible without the invaluable logistical help provided by Björn Björnsson of Nordfjördur. M.G.F. undertook the laboratory work on the lavas for an undergraduate dissertation and we have appreciated useful discussions with colleagues at Liverpool.

REFERENCES

Blakely, R.J., 1974. Geomagnetic reversals and crustal spreading rates during the Miocene. J. Geophys. Res., 79: 2979—2985.

Bodvarsson, G. and Walker, G.P.L., 1964. Crustal drift in Iceland. Geophys. J. R. Astron. Soc., 8: 285—300.

Charlton, S.R., 1973. An Investigation of Gases Released from Rocks on Fusion, with particular reference to potassium—argon dates. Ph.D. thesis, University of Liverpool.

Dagley, P., Wilson, R.L., Ade-Hall, J.M., Walker, G.P.L., Haggerty, S.E., Sigurgeirsson, T., Watkins, N.D., Smith, P.J., Edwards, J. and Grasty, R.L., 1967. Geomagnetic polarity zones for Icelandic lavas, Nature, 216: 25—29.

Doell, R.R. and Cox, A., 1967. Palaeomagnetic sampling with a portable drill. In: D.W. Collinson, K.M. Creer and S.K. Runcorn (editors), Methods in Palaeomagnetism. Elsevier, Amsterdam, pp. 21—25.

Gibson, I.L., 1966. The crustal structure of eastern Iceland. Geophys. J. R. Astron. Soc., 12: 99—102.

Gibson, I.L., 1969. A comparative account of the flood basalt volcanism of the Columbia Plateau and eastern Iceland. Bull Volcanol., 33: 419—437.

Gibson, I.L. and Piper, J.D.A., 1972. Structure of the Icelandic basalt plateau and the process of drift. Philos. Trans. R. Soc. Lond., A 271: 141—150.

Gibson, I.L., Kinsman, D.J.J. and Walker, G.P.L., 1966. Geology of the Faskrudsfjördur area, Eastern Iceland. Visindafélag Islandinga (Greinar IV), 2: 1—52.

Harrison, C.G.A., 1968. Formation of magnetic anomaly pattern by dike injection. J. Geophys. Res., 73: 2137—2142.

Heirtzler, J.R., Dickson, G.O., Herron, E.M., Pitman, W.C. and Le Pichon, X., 1968. Marine magnetic anomalies, geomagnetic reversals, and motions of the ocean floor and continents. J. Geophys. Res., 73: 2119–2136.

Khramov, A.N. and Sholpo, L.Ye, 1967. Palaeomagnetism, Nedra Press, Leningrad, pp. 213–233.

Klitgord, K.D., Huestis, S.P., Mudie, J.D. and Parker, R.L., 1975. An analysis of near-bottom magnetic anomalies: sea-floor spreading and the magnetised layer. Geophys. J. R. Astron. Soc., 43: 387–424.

Larson, E.E. and Speiss, F.N., 1969. The East Pacific Rise Crest: A near bottom geophysical profile. Science, 163: 68–71.

Marsh, B.D. and Carmichael, I.S.E., 1974. Benioff zone magmatism. J. Geophys. Res., 79: 1196–1204.

Matthews, D.H. and Bath, J., 1967. Formation of magnetic anomaly pattern of mid-Atlantic ridge. Geophys. J. R. Astron. Soc., 13: 349–357.

Moorbath, S., Sigurdsson, H. and Goodwin, R., 1968. K–Ar ages of the oldest exposed rocks in Iceland. Earth Planet. Sci. Lett., 4: 197–205.

Opdyke, N.D. and Henry, K.W., 1969. A test of the dipole hypothesis. Earth Planet. Sci. Lett., 6: 139–151.

Pálmason, G., 1973. Kinematics and heat flow in a volcanic rift zone with application to Iceland. Geophys. J. R. Astron. Soc., 33: 451–481.

Pálmason, G. and Saemundsson, K., 1974. Iceland in relation to the mid-Atlantic ridge. Ann. Rev. Earth Planet. Sci., 2: 25–50.

Piper, J.D.A., 1973a. Interpretation of some magnetic anomalies over Iceland. Tectonophysics, 16: 163–187.

Piper, J.D.A., 1973b. Volcanic history and tectonics of the north Langjökull region, central Iceland. Can. J. Earth Sci., 10: 164–179.

Ross, J.G. and Mussett, A.E., 1976. $^{40}Ar/^{39}Ar$ dates for spreading rates in eastern Iceland. Nature, 259: 36–38.

Saemundsson, K., 1967. Vulkanismus und Tektonik des Hengill-Gebietes, Acta Nat. Island, II, 7: 101 pp.

Serson, P.H., Hannaford, W.L. and Haines, G.V., 1968. Magnetic anomalies over Iceland. Science, 162: 355–357.

Vogt, P.R., 1974. Volcano height and plate thickness. Earth Planet. Sci. Lett., 23: 337–348.

Vine, F.J., 1966. Spreading of the ocean floor: new evidence. Science, 154: 1405–1415.

Walker, G.P.L., 1959. Geology of the Reydarfjördur area, eastern Iceland. Q. J. Geol. Soc. Lond., 114: 367–393.

Walker, G.P.L., 1960. Zeolite zones and dike distribution in relation to structure of basalts in eastern Iceland. J. Geol., 68: 515–528.

Walker, G.P.L., 1964. Geological investigations in eastern Iceland. Bull. Volcanol., 27: 351–366.

Walker, G.P.L., 1965. Evidence of crustal drift from Icelandic geology. Philos. Trans. R. Soc. Lond., A 258: 199–204.

Walker, G.P.L., 1975. Excess spreading axes and spreading rate in Iceland. Nature, 255: 468–470.

Watkins, N.D., Walker, G.P.L., Wilson, R.L. and Kristjansson, L., 1976. Magnetostratigraphy of eastern Iceland. Am. J. Sci., in press.

Wilson, R.L., 1970. Permanent aspects of the Earth's non-dipole magnetic field over Upper Tertiary times. Geophys. J. R. Astron. Soc., 19: 417–437.

Wilson, R.L., 1971. Dipole offset — the time average palaeomagnetic field over the past 25 million years. Geophys. J. R. Astron. Soc., 22: 491–504.

Wilson, R.L., 1972. Geomagnetic differences between normal and reversed field sources, and the problem of far-sided and right-handed pole positions, Geophys. J. R. Astron. Soc., 28: 295–304.